MAROKA MUSA CALVIN

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PROFESSIONAL SUMMARY

JUNIOUR AI ENGINEER | DATA SCIENCE

Computer Science Honours candidate and aspiring AI/ML Engineer with hands-on experience in machine learning, data analysis, deep learning, and backend development. Skilled in Python, TensorFlow/Keras, PyTorch, scikit-learn, Pandas, and NumPy, with practical experience building, training, evaluating, and documenting machine learning models. Experienced in applying data preprocessing, model evaluation, transfer learning, and end-to-end ML workflows to real-world projects. Seeking an entry-level opportunity where I can contribute to AI, machine learning, data science, or software engineering projects while continuing to develop advanced technical expertise.

TECHNICAL SKILLS

Languages:

Python

Databases:

SQL

Machine Learning & AI:

Machine Learning, Deep Learning, PyTorch, TensorFlow, Keras, Hugging Face, Scikit-learn

Data Science & Analysis:

Pandas, NumPy, Matplotlib, ScikitLearn/Seaborn, Data Cleaning, Data Preprocessing, Data Analysis

Development Tools: Git, GitHub, VS Code, Anaconda, PyCharm

Machine Learning Workflow:

Model Training, Model Evaluation, CI/CD for ML, MLOps Fundamentals

PROFESSIONAL EXPERIENCE

Machine Learning Intern | *GitHub*
Codveda Technologies | Remote
Aug 2026 – Sep 2026

- Completed practical machine learning tasks covering data preprocessing, regression, classification, clustering, ensemble learning, support vector machines, and neural networks.
- Cleaned and prepared datasets by handling missing values, encoding categorical variables, scaling numerical features, and splitting data into training and testing sets.
- Built and evaluated supervised machine learning models including Linear Regression, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine models.

- Applied model evaluation techniques including Mean Squared Error, R-squared, accuracy, precision, recall, F1-score, confusion matrix, cross-validation, ROC curve, and AUC.
- Performed feature importance analysis, model interpretation, and result visualization to explain model performance clearly.
- Developed and documented machine learning workflows using Python, pandas, NumPy, scikit-learn, matplotlib, seaborn, TensorFlow/Keras, Git, GitHub, and VS Code.

Aug 2026 | Python for Data Science, AI & Development – IBM | ***credential***

- Developed practical Python skills using core data structures and programming concepts for data-driven applications.
- Worked with Pandas and NumPy for data manipulation, cleaning, wrangling, and analysis.
- Used REST APIs and HTTP requests to retrieve and process external data.
- Practiced data collection and web scraping using BeautifulSoup and Selenium.
- Strengthened practical foundations for data science, machine learning, and backend development.

Sep 2026 | Generative AI and LLMs: Architecture and Data Preparation — IBM | *credential*

- Built practical knowledge of Generative AI architectures, including GANs, VAEs, diffusion models, and transformers.
- Learned how Large Language Models such as GPT and BERT are used in NLP-based applications.
- Worked with LLM data preparation concepts, including tokenization, data loaders, and model-ready text input.
- Gained exposure to Hugging Face Transformers and PyTorch for developing Generative AI and NLP applications.
- Learned how to identify AI hallucinations and evaluate model outputs for factual accuracy in real-world applications.

Sep 2026 | IBM Deep Learning with TensorFlow and Keras — ***Credential***

- Developed a binary image-classification model to distinguish between organic and recyclable waste using TensorFlow/Keras and a pretrained VGG16 convolutional neural network.
- Prepared and augmented image data using Keras ImageDataGenerator, including rescaling, image shifting and horizontal flipping, with separate training, validation and test generators.
- Implemented transfer learning and feature extraction by freezing pretrained VGG16 layers and adding custom dense/dropout classification layers, followed by fine-tuning of the later convolutional layers.
- Applied binary cross-entropy, Adam/RMSprop optimization, learning-rate scheduling, early stopping and model checkpointing, and visualized training/validation accuracy and loss curves.
- Evaluated model performance using precision, recall, F1-score and accuracy, achieving 80% accuracy with the feature-extraction model on the evaluation subset.

Tech: Python, TensorFlow, Keras, VGG16, NumPy, Matplotlib, scikit-learn, Computer Vision, Transfer Learning, CNNs

PROJECTS

Feb-Sep 2026 | End-to-End Speech Recognition (ML Project) | *GitHub*

Developed a machine learning model for speech recognition | Honours Research Project

- Fine-tuned the pretrained XLS-R 300M model using Hugging Face Transformers.
- Trained and evaluated the model using the NCHLT Northern Sotho speech dataset.
- Applied audio preprocessing, transcript normalization, and CTC-based model fine-tuning.
- Evaluated performance using Word Error Rate (WER), Character Error Rate (CER), and real-time factor analysis.
- Achieved a test WER of 22.78% and CER of 5.86% on unseen speech data.
- Demonstrated the effectiveness of transfer learning for low-resource African language speech recognition.

Tech: Python, PyTorch, Hugging Face Transformers, Librosa, Pandas

Sep 2026 | Waste Classification Using Transfer Learning — TensorFlow/Keras | *GitHub*

- Built a binary image-classification system for organic vs recyclable waste using a pretrained VGG16 CNN and ImageNet weights.
- Implemented image preprocessing, data augmentation, feature extraction and fine-tuning of VGG16 convolutional layers using TensorFlow/Keras.
- Applied dropout, learning-rate scheduling, early stopping and model checkpointing, with performance evaluation using accuracy, precision, recall and F1-score.
- Achieved 80% accuracy on the evaluation subset using the VGG16 feature-extraction approach.

Technologies: Python, TensorFlow, Keras, VGG16, NumPy, Matplotlib, scikit-learn

CERTIFICATIONS

Python for Machine Learning & Data Science Masterclass — Udemy, 2026 | *Live link*

Completed training in Python data science and machine learning, covering real-world datasets, feature engineering, classification, regression, data visualization, NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn.

EDUCATION

BSc in Mathematics and Computer Science | University of Limpopo

2023 – 2025

Honours Degree in Computer Science | University of Limpopo

Jan 2026 – Nov 2026

KEY STRENGTH

- **Machine Learning & Deep Learning Development** — Model development, training, evaluation, and optimization using Python and modern ML frameworks.
- **Data Analysis & Preprocessing** — Data cleaning, transformation, feature preparation, exploratory analysis, and visualization.
- **Computer Vision & Transfer Learning** — Image classification using CNNs, VGG16, TensorFlow, and Keras.
- **Backend & API Development** — Building backend systems, REST APIs, database-driven applications, and software solutions.
- **Problem Solving & System Thinking** — Breaking complex problems into practical, structured, and implementable solutions.
- **End-to-End Project Development** — Experience taking projects from data preparation and model development through evaluation, documentation, and version control.
- **Continuous Learning & Adaptability** — Quickly learning new technologies, frameworks, and development approaches.